

CONTINUUM LIMIT OF A ONE-DIMENSIONAL ATOMISTIC MODEL FOR MOIRÉ RELAXATION

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1. INTRODUCTION

Moiré materials are formed by stacking two-dimensional crystalline layers with a small relative twist angle or lattice mismatch. The superposition of the lattice periodicities produces a long-wavelength moiré pattern whose length scale can be much larger than the lattice spacing of either layer. Moiré materials have emerged as an ideal experimental platform for investigating strongly correlated electronic phases. Notable examples include superconductivity in magic-angle twisted bilayer graphene [1] and the fractional quantum anomalous Hall effect in twisted bilayer MoTe_2 [10].

Mechanical relaxation is the process by which atoms in moiré materials move from their monolayer equilibrium positions in order to minimize the total energy of the bilayer. This relaxation is governed by a competition between the elastic cost of deforming each layer and the dependence of the interlayer energy on local stacking. When the twist angle or lattice mismatch is small, this competition tends to enlarge regions of energetically favorable stacking and localize the transitions between them to narrow domain walls [9, 2, 3]. Atomic and electronic reconstruction for relaxation have been observed experimentally [13]. Together with calculations showing that relaxation can substantially modify the low-energy electronic band structure [9], these observations indicate that accurate prediction of moiré materials' electronic properties requires reliable modeling of the relaxed atomic structure.

A variety of approaches have therefore been used to determine relaxed structures, including density-functional theory calculations [11], molecular dynamics [12], multiscale modeling [6], and effective continuum theories [9, 2]. Common continuum models involve minimizing an energy that balances intralayer elasticity against an interlayer misfit energy determined by local stacking, using continuous displacement or modulation fields defined in real or configuration space [6, 9, 2, 4, 3]. Despite the widespread use of such continuum models, their precise relation to the underlying atomistic problem remains incompletely understood. In particular, Cazeaux, Luskin, and Massatt [4] discuss the question of whether solutions of their continuum elastostatic model are close to solutions of the atomistic problem that it approximates. This atomistic-to-continuum question motivates the present work.

To address this question, we study a one-dimensional minimal model. The main contribution of the present work is a microscopic derivation of the associated continuum relaxation functional, together with a quantitative atomistic-to-continuum convergence result. Specifically, we consider two periodic chains with common period 2π , containing N and $N+1$ atoms per period, respectively, so that the lattice mismatch tends to zero as $N \rightarrow \infty$. After rescaling, the atomistic energy consists of harmonic nearest-neighbor intralayer terms and an interlayer interaction

generated by an even, sufficiently regular pair potential V with polynomial decay. In the continuum limit, the interlayer interactions generate a periodic misfit potential by summing V over lattice translates, while the intralayer interactions produce the elastic energy. The resulting continuum relaxation functional is the one-dimensional analogue of the continuum models used in [9, 2]. Under suitable structural and stability assumptions, we prove that an atomistic minimizer u_d converges to the corresponding sampled continuum minimizer u_c in the discrete H^2 and maximum norms at rate $O(N^{-1})$. This provides a quantitative justification of the continuum model for the present atomistic system.

The paper is organized as follows. Section 2 analyzes the continuum energy, characterizing its symmetries, establishing existence and regularity of minimizers and proving a continuum H^2 -stability estimate. Section 3 introduces and analyzes the discrete energy, including its symmetries, existence of minimizers, Euler–Lagrange equations, and a discrete H^2 -stability estimate. Section 4 establishes consistency of the sampled continuum minimizer with the discrete Euler–Lagrange equations and then proves the discrete-to-continuum convergence theorem. The appendices contain the detailed derivation of the simplified atomistic model and proofs of the auxiliary estimates used in the discrete analysis.

2. WELL-POSEDNESS OF THE CONTINUUM ENERGY AND PROPERTIES OF MINIMIZERS

We consider the following continuum relaxation functional

$$(1) \quad E[u_1, u_2] = \int_{-\pi}^{\pi} \frac{1}{2} u_1'(x)^2 + \frac{1}{2} u_2'(x)^2 + \Phi(x + u_1(x) - u_2(x)) \, dx,$$

where $u_1, u_2 \in H_{\text{per}}^1 := \{u \in H_{\text{loc}}^1(\mathbb{R}) : u(x + 2\pi) = u(x)\}$ represent the displacements of the two layers. Without loss of generality, we identify functions $u \in H_{\text{per}}^1$ with their restriction to the fundamental cell $[-\pi, \pi]$ with periodic boundary conditions $u(\pi) = u(-\pi)$. We can impose periodicity conditions pointwise since $H^1(-\pi, \pi) \hookrightarrow C([-\pi, \pi])$.

The potential $\Phi \in C^{1,1}(\mathbb{R})$ which describes the interlayer misfit, is assumed to be even, 2π -periodic and satisfies

$$(2) \quad s \mapsto \frac{\Phi'(s)}{s} \text{ strictly increasing on } [0, \pi].$$

In particular, Φ is strictly decreasing on $(0, \pi)$ and attains its global minimum at $x = \pi$ modulo 2π . A canonical example to keep in mind is the cosine potential $\Phi(x) = \cos(x)$.

It is convenient to introduce the variables

$$(3) \quad u_+(x) := u_1(x) + u_2(x), \quad v(x) := x + u_1(x) - u_2(x).$$

Equivalently,

$$u_1(x) = \frac{1}{2}(u_+(x) + v(x) - x), \quad u_2(x) = \frac{1}{2}(u_+(x) - v(x) + x).$$

A direct computation gives

$$(4) \quad E[u_1, u_2] = \int_{-\pi}^{\pi} \frac{1}{4} u_+'(x)^2 \, dx + \int_{-\pi}^{\pi} \frac{1}{4} (v'(x) - 1)^2 + \Phi(v(x)) \, dx.$$

Since $u_1, u_2 \in H_{\text{per}}^1$, we have $u_+ \in H_{\text{per}}^1$ and $v \in x + H_{\text{per}}^1$ defined by

$$(5) \quad x + H_{\text{per}}^1 = \{x + u(x) : u \in H_{\text{per}}^1\}.$$

Note that we have $v(x+2\pi) = v(x) + 2\pi$ for $v \in x + H_{\text{per}}^1$. Again, we can equivalently consider functions $v \in H^1(-\pi, \pi)$ with the boundary condition $v(\pi) - v(-\pi) = 2\pi$. The first term of (4) is minimized by any constant $u_+ = C$. Thus, the minimization problem for $E[u_1, u_2]$ reduces entirely to the v part. Define $W := 2\Phi$, we have that

$$E[u_1, u_2] = \frac{1}{4} \int_{-\pi}^{\pi} u'_+(x)^2 dx + \frac{1}{2} J[v] - \frac{\pi}{2},$$

where

$$(6) \quad J[v] := \int_{-\pi}^{\pi} \frac{1}{2} v'(x)^2 + W(v(x)) dx.$$

Then minimizing (4) is equivalent to minimizing (6) over $v \in x + H_{\text{per}}^1$. First observe that J depends only on v' and on W , which are both 2π -periodic. Therefore J is invariant under vertical shifts by $2\pi\mathbb{Z}$:

$$J[v + 2\pi k] = J[v] \quad \text{for all } v \in x + H_{\text{per}}^1, k \in \mathbb{Z}.$$

For $d \in \mathbb{R}$ define translation $(\tau_d v)(x) := v(x - d)$. By change of variables we obtain

$$J[\tau_d v] = J[v].$$

In particular, if v_0 is a global minimizer, then so is $\tau_d v_0 + 2\pi k$ for every $d \in \mathbb{R}$ and $k \in \mathbb{Z}$. We will prove the following theorem on minimizers of (6) in $x + H_{\text{per}}^1$.

Theorem 1. *There exists a unique global minimizer $v_0 \in x + H_{\text{per}}^1$ of (6) satisfying $v_0(0) = 0$. All other minimizers have the form $v = \tau_d v_0 + 2\pi k$ for $d \in \mathbb{R}, k \in \mathbb{Z}$. In other words,*

$$\operatorname{argmin}_{x + H_{\text{per}}^1} J = \{v_0(x - d) - 2\pi k : d \in \mathbb{R}, k \in \mathbb{Z}\}.$$

Moreover, $v_0 \in C^{2,1}(\mathbb{R})$ is odd and satisfies $v'_0 > 0$.

The proof of Theorem 1 requires several ingredients which are collected below. First, the existence and regularity of a minimizer is a standard application of the direct method in the calculus of variations. We establish monotonicity in Proposition 1 and uniqueness in Proposition 2; the arguments are more involved but standard [7]. Finally, we establish stability of the Euler–Lagrange equation in H^2 norm in Proposition 3.

Proof of Theorem 1. For the existence, let $v_n = x + u_n \in x + H_{\text{per}}^1(-\pi, \pi)$ be a minimizing sequence. Using the symmetry $J[v + 2\pi k] = J[v]$, we can arrange it so that the means (u_n) are uniformly bounded. Now applying the Poincaré–Wirtinger inequality yields a uniform H^1 bound on u_n . Following the direct method as in [5, §3.4.1, Thm. 3.30], $\inf_{x + H_{\text{per}}^1} J$ attains its minimum.

For the regularity, we claim that every global minimizer v of J on $x + H_{\text{per}}^1$ is a $C^{2,1}$ strong solution of the Euler–Lagrange equation

$$(7) \quad v''(x) = W'(v(x)), \quad v(x + 2\pi) = v(x) + 2\pi.$$

Indeed, by Evans' Euler–Lagrange theorem for minimizers [8, Ch. 8, Thm. 4] any minimizer v satisfies (7) in the weak sense. Following the standard ODE theory, we can bootstrap directly to obtain $v \in C^{2,1}(\mathbb{R})$ since $W \in C^{1,1}(\mathbb{R})$.

Now let $v \in x + H_{\text{per}}^1$ be any global minimizer. By the regularity result, $v \in C^{2,1}(\mathbb{R})$, the endpoint values $v(\pm\pi)$ are well-defined. Choose $k \in \mathbb{Z}$ such that

$$v(-\pi) - 2\pi k \leq 0 < v(\pi) - 2\pi k.$$

By continuity, there exists $x_0 \in [-\pi, \pi]$ with $v(x_0) - 2\pi k = 0$. Define

$$v_0 := \tau_{-x_0} v - 2\pi k, \quad \text{so that} \quad v_0(0) = 0.$$

By the symmetry of J , v_0 is again a global minimizer in $x + H_{\text{per}}^1$ satisfying (7). Now define $w(x) := -v_0(-x)$. A direct computation gives

$$w''(x) = -v_0''(-x) = -W'(v_0(-x)) = W'(-v_0(-x)) = W'(w(x)),$$

so w satisfies the same ODE as v_0 on $(-\pi, \pi)$. Moreover,

$$w(0) = 0 = v_0(0), \quad w'(0) = v_0'(0).$$

The uniqueness for the ODE initial value problem gives $w = v_0$ on $[-\pi, \pi]$. Therefore v_0 is odd. Combining oddness with $v_0(\pi) - v_0(-\pi) = 2\pi$, we obtain

$$v_0(\pi) = \pi, \quad v_0(-\pi) = -\pi.$$

Thus every global minimizer is related by symmetries of the functional to an odd global minimizer satisfying $v_0(0) = 0$ and $v_0(\pm\pi) = \pm\pi$. Monotonicity and uniqueness of v_0 follow from Propositions 1 and 2 below. $\square$

Proposition 1 (Monotonicity). *Let $v \in x + H_{\text{per}}^1$ be an odd function that minimizes (6) and satisfies $v(0) = 0$. Then $v'(x) > 0$ for $-\pi \leq x \leq \pi$.*

Proof. Since $W \in C^{1,1}(\mathbb{R})$ is even, 2π -periodic with $W'(x) < 0$ for $x \in (0, \pi)$, we have $W'(\pi) = 0$ and $W(\pi) \leq W(x)$ for all $x \in \mathbb{R}$. As a minimizer, v satisfies (7) with the boundary conditions $v(\pm\pi) = \pm\pi$. Multiplying the ODE by v' , we obtain

$$\frac{1}{2} \frac{d}{dx} v'(x)^2 = v''(x)v'(x) = W'(v(x))v'(x) = \frac{d}{dx} W(v(x)) \quad x \in (-\pi, \pi)$$

Integrating both sides from x to π , we get that

$$v'(x)^2 = v'(\pi)^2 + 2(W(v(x)) - W(\pi)) \geq v'(\pi)^2 \quad \forall x \in [-\pi, \pi],$$

where the inequality uses $W(v(x)) \geq W(\pi)$.

If $v'(\pi) = 0$, then the constant function $x \mapsto \pi$ solves $v'' = W'(v)$ and has the same Cauchy data $(v(\pi), v'(\pi)) = (\pi, 0)$; by uniqueness for the ODE, this would force $v \equiv \pi$, contradicting $v(0) = 0$. Thus $v'(\pi) \neq 0$, and therefore $v'(x)^2 \geq v'(\pi)^2 > 0$ on $[-\pi, \pi]$. In particular, v' never vanishes and hence has a fixed sign. Since $v(\pi) - v(-\pi) = 2\pi > 0$, this sign must be positive, and so $v'(x) > 0$ for all $x \in [-\pi, \pi]$. $\square$

Proposition 2 (Uniqueness). *There exists a unique minimizer v satisfying $v(0) = 0$.*

Proof. Let v_1 and v_2 be two global minimizers of J in $x + H_{\text{per}}^1$ such that $v_i(0) = 0$, satisfying (7) with the boundary conditions $v_i(\pm\pi) = \pm\pi$ for $i = 1, 2$. Proposition 1 yields

$$(8) \quad v_i'(x) > 0 \quad \text{for all } x \in [-\pi, \pi].$$

In particular, restricting to $[0, \pi]$, each v_i satisfies

$$(9) \quad v'' = W'(v), \quad v(0) = 0, \quad v(\pi) = \pi, \quad v'(x) > 0 \text{ on } (0, \pi).$$

We claim that (9) has at most one solution.

Note that if $v'_1(\pi) = v'_2(\pi)$, then $(v_1(\pi), v'_1(\pi)) = (v_2(\pi), v'_2(\pi)) = (\pi, v'_1(\pi))$, and uniqueness for the ODE initial value problem implies $v_1 \equiv v_2$ on $[0, \pi]$. Thus we assume $v'_1(\pi) \neq v'_2(\pi)$ and, without loss of generality, $v'_1(\pi) < v'_2(\pi)$.

Since $v_1(\pi) = v_2(\pi) = \pi$ and both are increasing, there exists $\delta > 0$ such that $v_1(x) > v_2(x)$ for all $x \in (\pi - \delta, \pi)$. Let (a, π) be the largest interval on which $v_1 > v_2$, where $a \in [0, \pi)$. Then $v_1(a) = v_2(a) =: b \in [0, \pi)$, $v_1 > v_2$ on (a, π) and $v'_1(a) \geq v'_2(a)$. Using $v''_i = W'(v_i)$, we have:

$$\int_a^\pi (v'_1 v'_2 + v_1 W'(v_2)) dx = \int_a^\pi \frac{d}{dx} (v_1 v'_2) dx = \pi v'_2(\pi) - b v'_2(a).$$

Swapping v_1 and v_2 above and subtracting yields

$$\int_a^\pi (v_2 W'(v_1) - v_1 W'(v_2)) dx = \pi (v'_1(\pi) - v'_2(\pi)) - b (v'_1(a) - v'_2(a)) < 0.$$

On the other hand, for $x \in [a, \pi]$, $v_1(x) \geq v_2(x) \geq 0$, using (2) we have

$$v_2 W'(v_1) - v_1 W'(v_2) = v_1 v_2 \left(\frac{W'(v_1)}{v_1} - \frac{W'(v_2)}{v_2} \right) \geq 0 \quad \text{for all } x \in [a, \pi],$$

a contradiction. Therefore $v_1 \equiv v_2$ on $[0, \pi]$.

Finally, since both minimizers are odd, equality on $[0, \pi]$ implies equality on $[-\pi, 0]$. Thus $v_1 \equiv v_2$ on $[-\pi, \pi]$, proving the uniqueness of the minimizer with $v(0) = 0$. $\square$

In addition to Theorem 1, we claim the Euler–Lagrange equation (7) is stable in H^2 in the sense of the following Proposition 3. Note that here the stability requires the nonlinearity (i.e., the potential) to be small, while all the previous results up to here hold in general. Specifically, we require the Lipschitz constant of W' to be less than 1, which is the Poincaré constant for periodic functions on $(-\pi, \pi)$.

Proposition 3 (Stability). *Assume that $W \in C^{1,1}(\mathbb{R})$ with Lipschitz constant $L := \text{Lip}(W') < 1$. Let $v, w \in x + H^2_{\text{per}}$ and $r \in L^2(-\pi, \pi)$ satisfy*

$$v'' = W'(v), \quad w'' = W'(w) + r \quad \text{a.e. on } (-\pi, \pi).$$

Set $g := v - w$ and denote the mean by

$$(g) := \frac{1}{2\pi} \int_{-\pi}^\pi g(x) dx.$$

Then there exists a constant $C = C(L) > 0$ such that

$$\|g\|_{H^2(-\pi, \pi)} \leq C (\|r\|_{L^2(-\pi, \pi)} + |(g)|).$$

Proof. Note that $g'' = W'(v) - W'(w) - r$ and $g = v - w \in H^2_{\text{per}}$. The function $g - (g)$ is periodic with zero mean and g' has zero mean by periodicity. Applying Poincaré–Wirtinger inequality to $g - (g)$ and then to g' on $[-\pi, \pi]$, where the Poincaré constant is 1, then

$$\|g - (g)\|_{L^2} \leq \|g'\|_{L^2} \leq \|g''\|_{L^2},$$

and

$$\|g''\|_{L^2} \leq \|W'(v) - W'(w)\|_{L^2} + \|r\|_{L^2} \leq L \|g\|_{L^2} + \|r\|_{L^2}.$$

Using the triangle inequality

$$\|g\|_{L^2} \leq \|g - (g)\|_{L^2} + \sqrt{2\pi} |(g)| \leq \|g''\|_{L^2} + \sqrt{2\pi} |(g)|.$$

Combining the above estimates, we get

$$\|g''\|_{L^2} \leq \frac{L\sqrt{2\pi}}{1-L} |(g)| + \frac{1}{1-L} \|r\|_{L^2}$$

The same bound holds for $\|g - (g)\|_{L^2}$ and $\|g'\|_{L^2}$, so

$$\|g\|_{H^2(-\pi,\pi)} \leq C(L) (\|r\|_{L^2} + |(g)|).$$

□

3. WELL-POSEDNESS OF THE DISCRETE ENERGY

We consider a minimal atomic-scale model for relaxation in moiré materials: two one-dimensional chains of atoms with a small lattice mismatch analogous to a two-dimensional twist, where the atoms interact via pair potentials.

We consider the case where the system has an exact periodic cell given by $[0, 2\pi)$. Fix an integer $N \geq 2$, we denote the number of atoms in layer i within this cell by N_i with $N_1 = N$ and $N_2 = N + 1$. The lattice constants are thus $h_i = \frac{2\pi}{N_i}$, and the lattice mismatch θ is defined by

$$h_2 = (1 - \theta)h_1 \implies \theta = \frac{1}{N + 1}.$$

For $\xi = (\xi^1, \xi^2)$, let ξ_n^i denote the displacement of the n -th atom in layer i from its reference position nh_i . We define the atomic-scale energy by

$$(10) \quad \mathcal{E}_N[\xi] := \mathcal{E}^1[\xi] + \mathcal{E}^2[\xi],$$

where $V_{i,i}$ and $V_{i,3-i}$, $i \in \{1, 2\}$, are decaying pair potentials and

$$(11) \quad \begin{aligned} \mathcal{E}^i[\xi] := & \\ & h_i \sum_{n=0}^{N_i-1} \sum_{m \in \mathbb{Z}} [V_{i,i}(nh_i + \xi_n^i - mh_i - \xi_m^i) + V_{i,3-i}(nh_i + \xi_n^i - mh_{3-i} - \xi_m^{3-i})]. \end{aligned}$$

Note that the arguments of the potentials are the relative atomic positions, including both reference separation and displacement, and that the infinite sum over m accounts for the periodicity of the chains. Throughout, any discrete pair $u = (u^1, u^2)$ is understood to satisfy the periodicity condition

$$u_{n+N_i}^i = u_n^i.$$

Rather than considering (11) directly, we further simplify this model as follows. For layer i , define the forward and backward differences

$$\nabla_i^+ u_n^i := \frac{u_{n+1}^i - u_n^i}{h_i}, \quad \nabla_i^- u_n^i := \frac{u_n^i - u_{n-1}^i}{h_i},$$

and the periodic discrete Laplacian

$$\Delta_i u_n^i := \nabla_i^- \nabla_i^+ u_n^i = \frac{u_{n+1}^i - 2u_n^i + u_{n-1}^i}{h_i^2}.$$

Denote $\sum_{n=0}^{N_i-1} := h_i \sum_{n=0}^{N_i-1}$ and take $I_i = \{0, \dots, N_i - 1\}$, $h = \frac{2h_1 h_2}{h_1 + h_2}$. Then, after re-scaling the displacement functions $\xi^i = \frac{h_i}{2\pi} u^i$ and some further manipulations

(see Appendix A for a detailed derivation), the atomistic energy (10)-(11) can be simplified to

$$E_N[u] := E^1[u] + E^2[u],$$

where

$$(12) \quad E^i[u] := \frac{1}{2} \sum_{n \in I_i} (\nabla_i^+ u_n^i)^2 + \sum_{n \in I_i} \sum_{m \in \mathbb{Z}} V(A_{n,m}^i[u]), \quad i = 1, 2,$$

$$(13) \quad A_{n,m}^i[u] := (-1)^{3-i} h n + \frac{h}{h_{3-i}} u_n^i - \frac{h}{h_i} u_{n+m}^{3-i} - 2\pi \frac{h}{h_i} m, \quad n \in I_i.$$

We equip our system with the layerwise weighted ℓ^2 norm:

$$\|u^i\|_{\ell_i^2}^2 := h_i \sum_{n=0}^{N_i-1} |u_n^i|^2,$$

and the rescaled ℓ^2 norm:

$$\|u\|_{\ell_h^2}^2 := \|u^1\|_{\ell_1^2}^2 + \|u^2\|_{\ell_2^2}^2.$$

For the discrete derivatives, we write

$$\begin{aligned} \|\nabla_h u\|_{\ell_h^2}^2 &:= \|\nabla_1^+ u^1\|_{\ell_1^2}^2 + \|\nabla_2^+ u^2\|_{\ell_2^2}^2, \\ \|\Delta_h u\|_{\ell_h^2}^2 &:= \|\Delta_1 u^1\|_{\ell_1^2}^2 + \|\Delta_2 u^2\|_{\ell_2^2}^2. \end{aligned}$$

By periodicity, the forward and backward difference operators have the same layerwise ℓ_h^2 norm, so the choice of the forward difference in this definition is immaterial. We define the discrete H^2 norm by

$$\|u\|_{H_h^2}^2 := \|u\|_{\ell_h^2}^2 + \|\nabla_h u\|_{\ell_h^2}^2 + \|\Delta_h u\|_{\ell_h^2}^2.$$

Note that the additive quantity used below is an equivalent norm:

$$\|u\|_{H_h^2} \leq \|u\|_{\ell_h^2} + \|\nabla_h u\|_{\ell_h^2} + \|\Delta_h u\|_{\ell_h^2} \leq \sqrt{3} \|u\|_{H_h^2}.$$

We also use layerwise discrete maximum norm and the maximum norm for the two-layer system,

$$\|u^i\|_{\infty,i} := \max_{n \in I_i} |u_n^i|, \quad \|u\|_{\infty} := \max_{i=1,2} \|u^i\|_{\infty,i}.$$

Note that one has the estimate

$$\|u\|_{\infty} \leq \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2} \leq 2\|u\|_{\infty}.$$

For the layerwise means, we use

$$(u^i) := \frac{h_i}{2\pi} \sum_{n=0}^{N_i-1} u_n^i.$$

3.1. Existence of Minimizers. From this point on, we assume that the potential $V \in C^2(\mathbb{R})$ is even and satisfies the following polynomial decay condition: for some $r > 2$ and $C_V > 0$,

$$(14) \quad |V(x)| + (1 + |x|) |V'(x)| + (1 + |x|)^2 |V''(x)| \leq C_V (1 + |x|)^{-r}, \quad x \in \mathbb{R}.$$

From now on, we write $A \lesssim B$ to mean that $A \leq CB$ for a constant C independent of the discretization parameters N, h and of the lattice indices. The problem-dependent constants whose influence we track separately are the potential constant C_V , the discrete Poincaré constant C_P , and the force Lipschitz constant L_F introduced below; when their dependence matters, we keep it explicit.

Under these assumptions, and in particular using the evenness of V , the discrete energy (12) enjoys the following translation and reflection symmetries.

Lemma 1 (Discrete symmetries). *For $d \in \mathbb{Z}$ and $c \in \mathbb{R}$, define the operator*

$$(15) \quad (\tau_{d,c} u)_n^1 = u_{n+d}^1 + h_2 d + h_2 c, \quad (\tau_{d,c} u)_n^2 = u_{n+d}^2 + h_1 c.$$

Here all indices are understood modulo the corresponding layer period N_i . Then

$$E_N[\tau_{d,c} u] = E_N[u], \quad d \in \mathbb{Z}, \quad c \in \mathbb{R}.$$

Consequently, the layer means transform under $\tau_{d,c}$ by

$$((\tau_{d,c} u)^1) = (u^1) + h_2 d + h_2 c, \quad ((\tau_{d,c} u)^2) = (u^2) + h_1 c.$$

Moreover, even-ness of V implies the odd-reflection operator

$$(\mathcal{R}u)_n^i := -u_{-n}^i, \quad i = 1, 2,$$

is also a symmetry:

$$E_N[\mathcal{R}u] = E_N[u].$$

Use Lemma 1, we have the following corollary that allows us to bound the layer means of a configuration without changing its energy.

Corollary 1 (Mean-fixing by translation). *For every discrete pair $(u^1, u^2) \in \mathbb{R}^N \times \mathbb{R}^{N+1}$, there exists a discrete pair $(\tilde{u}^1, \tilde{u}^2)$ satisfying $E_N[\tilde{u}] = E_N[u]$ such that*

$$(\tilde{u}^1) + (\tilde{u}^2) = 0, \quad |(\tilde{u}^1)| = |(\tilde{u}^2)| \leq \frac{h}{4}.$$

We next record a near-far field estimate for the interaction term, obtained directly from the decay assumption (14).

Lemma 2 (Near-far field splitting). *For every $n \in I_i$ and $m \in \mathbb{Z}$, $A_{n,m}^i[u]$ defined by (13). Assume (14) holds. Then for any $G = V, V', V''$, we have*

$$\sum_{m \in \mathbb{Z}} \sup_{n \in I_i} |G(A_{n,m}^i[u])| \leq 13C_V \left(1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}\right).$$

We obtain an a priori upper bound for the energy of minimizers by testing (12) with the zero configuration $u_n^i = 0$. Indeed, Lemma 2 gives

$$E^i[0] \leq \sum_{n \in I_i} \sum_{m \in \mathbb{Z}} \sup_{n \in I_i} \left| V \left((-1)^{3-i} h n - 2\pi \frac{h}{h_i} m \right) \right| \leq 26\pi C_V.$$

In what follows we restrict attention to configurations satisfying $E_N[u] \leq C_0$, where $C_0 := \sup_{N \in \mathbb{N}} E_N[0] \leq 52\pi C_V$.

Theorem 2 (Existence of discrete minimizer). *Let*

$$\mathcal{A}_N := \left\{ (u^1, u^2) \in \mathbb{R}^N \times \mathbb{R}^{N+1} : (u^1) + (u^2) = 0, |(u^i)| \leq \frac{h}{4} \right\}.$$

There exists a $u_ \in \mathcal{A}_N$ such that*

$$E_N[u_*] = \inf_{\mathcal{A}_N} E_N[u] = \inf_{\mathbb{R}^N \times \mathbb{R}^{N+1}} E_N[u].$$

The proof is a finite-dimensional direct-method argument once the translation symmetry has been normalized. We first use Corollary 1 to choose a normalized representative in each symmetry class, and then use the zero configuration together with Lemma 2 to obtain a uniform bound on the normalized sublevel set (see Lemma 3). These two ingredients make the relevant sublevel set compact, so the energy attains its minimum there.

We first apply Corollary 1 to prove that every normalized state u is uniformly bounded given that $E_N[u] \leq C_0$.

Lemma 3 (Uniform boundedness for normalized states). *Define the sublevel set*

$$\mathcal{K}_N := \left\{ u \in \mathcal{A}_N : E_N[u] \leq C_0 \right\},$$

where $C_0 := \sup_{N \in \mathbb{N}} E_N[0] \leq 52\pi C_V$. Assume $u \in \mathcal{K}_N$, then

$$\|u^1\|_{\infty,1} + \|u^2\|_{\infty,2} \leq \frac{h}{2} + 42\pi\sqrt{C_V} + 416\pi^2 C_V.$$

Since $h = \frac{4\pi}{2N+1}$, the sum of the layerwise maximum norms can be bounded uniformly in N by relaxing the RHS as

$$\|u^1\|_{\infty,1} + \|u^2\|_{\infty,2} \leq 42\pi\sqrt{C_V} + 416\pi^2 C_V + \frac{2\pi}{3}.$$

Proof of Theorem 2. By the translation symmetry (15) and Corollary 1, we have

$$\inf_{\mathbb{R}^N \times \mathbb{R}^{N+1}} E_N = \inf_{\mathcal{A}_N} E_N.$$

Any minimizer over $\mathcal{A}_N$ must lie in the sublevel set $\mathcal{K}_N$ since the zero configuration is a competitor, i.e.

$$\inf_{\mathcal{A}_N} E_N = \inf_{\mathcal{K}_N} E_N.$$

The set $\mathcal{K}_N$ is nonempty and by Lemma 3 is bounded. Now for fixed N , the layer means are continuous and hence $\mathcal{A}_N$ is closed. E_N is continuous, hence $\mathcal{K}_N$ is also closed, and thus compact in the finite-dimensional space $\mathbb{R}^N \times \mathbb{R}^{N+1}$. Therefore E_N attains its minimum on $\mathcal{K}_N$, and hence also on $\mathbb{R}^N \times \mathbb{R}^{N+1}$. $\square$

3.2. Euler-Lagrange equation. Now take

$$(16) \quad F_n^i(u) := 2 \sum_{m \in \mathbb{Z}} V' \left(A_{n,m}^i[u] \right), \quad n \in I_i, \quad i = 1, 2.$$

Then the Euler-Lagrange equation of (12) can be written as

$$(17) \quad \Delta_i u^i = F^i(u), \quad i = 1, 2.$$

We next record a consequence of (17) for uniformly bounded states.

Lemma 4 (Discrete Lipschitz bound for u_n^i). *Assume $u = (u^1, u^2) \in \mathcal{K}_N$ solves the discrete Euler–Lagrange equation (17), then $\|\nabla_i^+ u^i\|_{\infty, i}$ is also uniformly bounded in N for $i = 1, 2$. Consequently, for every $n \in I_i$ and $m \in \mathbb{Z}$, we have the following discrete Lipschitz bound for u^i :*

$$|u_{n+m}^i - u_n^i| \leq 32448\pi^3 \left(C_V + C_V^{3/2} + C_V^2 \right) |m|h.$$

We use this discrete Lipschitz bound to compare the full interaction in (12) with a localized form (18). This localized form has an exact vertical-shift symmetry, which yields the following approximate symmetry for the original discrete energy.

Lemma 5 (Approximate symmetry). *Consider the localized energy functional with $i \in \{1, 2\}$:*

$$(18) \quad E_{\text{loc}}^i[u] := \frac{1}{2} \sum_{n \in I_i} (\nabla_i^+ u_n^i)^2 + \sum_{n \in I_i} \sum_{m \in \mathbb{Z}} V((-1)^{3-i} hn + u_n^i - u_n^{3-i} - 2\pi m),$$

which the following symmetry holds and keeps $E_{\text{loc}}^i[\tau_k u] = E_{\text{loc}}^i[u]$:

$$(19) \quad (\tau_k u)_n^1 = u_n^1 + 2\pi k_1, \quad (\tau_k u)_n^2 = u_n^2 + 2\pi k_2, \quad \forall k_1, k_2 \in \mathbb{Z}.$$

For $V \in C^2(\mathbb{R})$ satisfying (14), u uniformly bounded as in Lemma 3, and satisfying (17); define the polynomial for C_V as

$$(20) \quad \mathcal{P}(C_V) := 10^{12}\pi^8 \left(C_V + C_V^{3/2} + C_V^2 + C_V^{5/2} + C_V^3 + C_V^{7/2} + C_V^4 + C_V^{9/2} + C_V^5 \right).$$

then for $i = 1, 2$,

$$|E^i[u] - E_{\text{loc}}^i[u]| \leq h\mathcal{P}(C_V).$$

Consequently, for any fixed $k = (k_1, k_2) \in \mathbb{Z}^2$, we have

$$|E^i[\tau_k u] - E^i[u]| \lesssim h.$$

Remark 3.1. *This approximate symmetry is a discrete analogue of the $J[v + 2\pi k] = J[v]$ symmetry in the continuum energy. In the continuum limit, this holds exactly, but in the discrete model, it only holds approximately up to an error of order $O(h)$.*

3.3. Discrete H^2 stability. We now state the discrete stability estimate for the Euler–Lagrange equation (17). As in the continuum stability result, the estimate requires a smallness condition on the nonlinear term, formulated below in terms of the Lipschitz constant of the force map $F = (F^1, F^2)$.

Let v_0 be the normalized continuum solution of (7) satisfying $v_0(0) = 0$. The corresponding continuum pair is

$$u_1(x) = \frac{v_0(x) - x}{2}, \quad u_2(x) = \frac{x - v_0(x)}{2}.$$

We denote by $u_c = (u_c^1, u_c^2)$ the sampling of this continuum pair on the two grids, namely

$$(u_c^i)_n := u_i(x_n^i), \quad n \in I_i, \quad i = 1, 2.$$

Here x_n^i denotes the grid point in layer i . Since v_0 satisfies $-\pi \leq v_0(x) \leq \pi$ for all $x \in [-\pi, \pi]$, the periodic sampled pair u_c is uniformly bounded in $\mathbb{R}$:

$$(21) \quad \|u_c^1\|_{\infty, 1} + \|u_c^2\|_{\infty, 2} \leq 2\pi.$$

Let $u_d = (u_d^1, u_d^2)$ be an exact discrete solution of (17). In the argument below, u_d will be the symmetry-normalized discrete minimizer constructed in Corollary 1; in particular, by Lemma 3, u_d is also uniformly bounded by:

$$(22) \quad \|u_d^1\|_{\infty,1} + \|u_d^2\|_{\infty,2} \leq 42\pi\sqrt{C_V} + 416\pi^2 C_V + \frac{2\pi}{3}.$$

The sampled continuum pair u_c is not an exact solution of the discrete Euler–Lagrange equation. We define its residual by

$$(23) \quad r^i := \Delta_i u_c^i - F^i(u_c), \quad i = 1, 2.$$

The consistency estimate will later give $\|r\|_{\ell_h^2} = O(h)$.

Define the error

$$w = (w^1, w^2) := u_d - u_c.$$

Subtracting (23) from (17) gives

$$(24) \quad \Delta_i w^i = F^i(u_d) - F^i(u_c) - r^i, \quad i = 1, 2.$$

Theorem 3 (Discrete H^2 -stability). *Let u_c , u_d , r , and $w = u_d - u_c$ be defined as above. Let C_P be the Poincaré constant defined in Lemma 7, and let L_F be the Lipschitz constant of the force map $F = (F^1, F^2)$ defined in Lemma 6. Assume the smallness condition*

$$(25) \quad C_P L_F < 1,$$

with the gap $1 - C_P L_F$ bounded below independently of N . Then there exists a constant $C > 0$ depending only on C_P and L_F , such that the error w satisfies the discrete H^2 -stability estimate

$$(26) \quad \|w\|_{H_h^2} \leq \|w\|_{\ell_h^2} + \|\nabla_h w\|_{\ell_h^2} + \|\Delta_h w\|_{\ell_h^2} \leq C \left(\|r\|_{\ell_h^2} + (|(w^1)| + |(w^2)|) \right),$$

where

$$C = \frac{1}{1 - C_P L_F} \max \left\{ 1 + \sqrt{C_P} + C_P, \sqrt{2\pi} \left[1 + \left(1 + \sqrt{C_P} \right) L_F \right] \right\}.$$

The proof follows the error equation (24). The constants L_F and C_P appearing in the statement are defined in the two lemmas below. The first ingredient is a force estimate $F(u_d) - F(u_c)$, which uses the uniform boundedness of both states. The second ingredient is a two-layer discrete Poincaré inequality, which converts control of $\Delta_h w$ into control of $\nabla_h w$ and w , up to the layer means. The smallness condition $C_P L_F < 1$ then allows the ℓ_h^2 -term to be absorbed.

To state the force estimate, we connect the sampled continuum state u_c and the exact discrete state u_d by the line segment

$$z_t := u_c + t(u_d - u_c), \quad t \in [0, 1].$$

The following lemma is the point where the uniform boundedness results for u_c and u_d enter.

Lemma 6. *Assume that V'' satisfies the polynomial decay condition (14). Let u_c and u_d be the normalized states defined above. Then for $i = 1, 2$,*

$$(27) \quad \max_{i=1,2} \sup_{N \in \mathbb{N}} \sup_{n \in I_i} \sum_{m \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{n,m}^i[z_t])| \leq 13C_V \left(1 + \frac{8\pi}{3} + 42\pi C_V^{1/2} + 416\pi^2 C_V \right)$$

is uniformly bounded, and consequently, the force F admits a uniform Lipschitz bound with Lipschitz constant L_F :

$$(28) \quad \|F(u_d) - F(u_c)\|_{\ell_h^2} \leq L_F \|u_d - u_c\|_{\ell_h^2},$$

where

$$L_F := 4\sqrt{2} \max_{i=1,2} \sup_{N \in \mathbb{N}} \sup_{n \in I_i} \sum_{m \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{n,m}^i[z_t])|.$$

The remaining ingredient for the stability estimate is the following two-layer discrete Poincaré–Wirtinger inequality.

Lemma 7 (Two-layer discrete Poincaré–Wirtinger inequality). *Let $u = (u^1, u^2) \in \mathbb{R}^N \times \mathbb{R}^{N+1}$, with periodic conventions $u_{n+N_i}^i = u_n^i$. Then, for each $i = 1, 2$,*

$$(29) \quad \|u^i\|_{\ell_i^2}^2 \leq C_{P,i} \|\nabla_i^+ u^i\|_{\ell_i^2}^2 + 2\pi |(u^i)|^2, \quad C_{P,i} = \frac{h_i^2}{4 \sin^2(h_i/2)}$$

Consequently, with $C_P := C_{P,1}$, one obtains $1 \leq C_P \leq \frac{\pi^2}{4}$ for $N \geq 2$ and

$$\|u\|_{\ell_h^2}^2 \leq C_P \left(\|\nabla_1^+ u^1\|_{\ell_1^2}^2 + \|\nabla_2^+ u^2\|_{\ell_2^2}^2 \right) + 2\pi \left((u^1)^2 + (u^2)^2 \right).$$

Proof of Theorem 3. Take the product ℓ_h^2 -norm of the error equation (24) and use the force Lipschitz bound (28), one obtains

$$(30) \quad \|\Delta_h w\|_{\ell_h^2} \leq \|F(u_d) - F(u_c)\|_{\ell_h^2} + \|r\|_{\ell_h^2} \leq L_F \|w\|_{\ell_h^2} + \|r\|_{\ell_h^2}.$$

Next we apply the Poincaré inequality (29) to w^i and summing over the two layers gives

$$(31) \quad \|w\|_{\ell_h^2}^2 \leq C_P \|\nabla_h w\|_{\ell_h^2}^2 + 2\pi \left(|(w^1)|^2 + |(w^2)|^2 \right).$$

Since $\nabla_i^- w^i$ has zero mean by periodicity, another application of (29) gives

$$\|\nabla_i^- w^i\|_{\ell_i^2}^2 \leq C_{P,i} \|\nabla_i^+ \nabla_i^- w^i\|_{\ell_i^2}^2 = C_{P,i} \|\Delta_i w^i\|_{\ell_i^2}^2.$$

we obtain

$$(32) \quad \|\nabla_h w\|_{\ell_h^2}^2 \leq C_P \|\Delta_h w\|_{\ell_h^2}^2.$$

Combining (31) and (32), we get

$$\|w\|_{\ell_h^2}^2 \leq C_P^2 \|\Delta_h w\|_{\ell_h^2}^2 + 2\pi \left(|(w^1)|^2 + |(w^2)|^2 \right),$$

and hence

$$(33) \quad \|w\|_{\ell_h^2} \leq C_P \|\Delta_h w\|_{\ell_h^2} + \sqrt{2\pi} (|(w^1)| + |(w^2)|).$$

Substituting (30) into (33) yields

$$\|w\|_{\ell_h^2} \leq C_P L_F \|w\|_{\ell_h^2} + C_P \|r\|_{\ell_h^2} + \sqrt{2\pi} (|(w^1)| + |(w^2)|).$$

By the smallness condition (25), we can absorb the first term on the right-hand side and obtain

$$(34) \quad \|w\|_{\ell_h^2} \leq \frac{C_P}{1 - C_P L_F} \|r\|_{\ell_h^2} + \frac{\sqrt{2\pi}}{1 - C_P L_F} (|(w^1)| + |(w^2)|).$$

From (30)

$$(35) \quad \|\Delta_h w\|_{\ell_h^2} \leq \frac{1}{1 - C_P L_F} \|r\|_{\ell_h^2} + \frac{\sqrt{2\pi} L_F}{1 - C_P L_F} (|(w^1)| + |(w^2)|).$$

From (32)

$$(36) \quad \|\nabla_h w\|_{\ell_h^2} \leq \frac{\sqrt{C_P}}{1 - C_P L_F} \|r\|_{\ell_h^2} + \frac{\sqrt{2\pi C_P} L_F}{1 - C_P L_F} (|(w^1)| + |(w^2)|).$$

Adding the three estimates gives

$$\begin{aligned} \|w\|_{\ell_h^2} + \|\nabla_h w\|_{\ell_h^2} + \|\Delta_h w\|_{\ell_h^2} &\leq \frac{1 + \sqrt{C_P} + C_P}{1 - C_P L_F} \|r\|_{\ell_h^2} \\ &\quad + \frac{\sqrt{2\pi} [1 + (1 + \sqrt{C_P}) L_F]}{1 - C_P L_F} (|(w^1)| + |(w^2)|). \end{aligned}$$

By taking C as

$$C = \frac{1}{1 - C_P L_F} \max \left\{ 1 + \sqrt{C_P} + C_P, \sqrt{2\pi} [1 + (1 + \sqrt{C_P}) L_F] \right\},$$

we obtain the desired H^2 -stability estimate (26). $\square$

4. CONSISTENCY OF THE EULER–LAGRANGE EQUATION AND CONVERGENCE

We now show that the sampled continuum solution u_c is an approximate solution of the discrete Euler–Lagrange equation. Throughout this section we assume that the continuum potential Φ is generated by the atomistic potential V through the periodization

$$(37) \quad \Phi(s) := 2 \sum_{m \in \mathbb{Z}} V(s - 2\pi m).$$

The factor 2 comes from the two layer contributions in the discrete energy.

Theorem 4 (Consistency of the discrete Euler–Lagrange equation). *Let v_0 be the normalized continuum solution of (7) with $v_0(0) = 0$, and let $u_c = (u_c^1, u_c^2)$ be its sampling on the two grids. Let r be the residual defined by (23). Then*

$$(38) \quad \|r\|_{\infty} \leq 1300(C_V^{3/2} + C_V)h,$$

and

$$(39) \quad \|r\|_{\ell_h^2} \leq 5000(C_V^{3/2} + C_V)h.$$

Proof. First by the decay assumption (14) and the periodization (37), the series for Φ, Φ', Φ'' converge uniformly and one obtains a convenient relaxed bound as:

$$(40) \quad \|\Phi\|_{L^\infty}, \|\Phi'\|_{L^\infty}, \|\Phi''\|_{L^\infty} \leq 4C_V.$$

Now use the first integral for (7):

$$(v'_0(x))^2 = (v'_0(\pi))^2 + 4(\Phi(v_0(x)) - \Phi(\pi)).$$

Since $v'_0 > 0$, $v'_0(\pi)$ is its minimum in $[-\pi, \pi]$, while

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} v'_0(x) dx = 1$$

so $v'_0(\pi) \leq 1$. Consequently,

$$(41) \quad \|v'_0\|_{L^\infty}^2 \leq 1 + 8\|\Phi\|_{L^\infty} \leq 1 + 32C_V.$$

Recall that the continuum pair is defined by

$$u_1(x) = \frac{v_0(x) - x}{2}, \quad u_2(x) = \frac{x - v_0(x)}{2},$$

so that

$$v_0(x) = x + u_1(x) - u_2(x).$$

Apply the above estimate (41) one obtains

$$(42) \quad \|(u_i)'\|_{L^\infty} \leq 3 \left(\sqrt{C_V} + 1 \right).$$

Since v_0 solves (7), the continuum pair satisfies

$$(43) \quad (u_i)''(x) = (-1)^{3-i} \Phi'(v_0(x)), \quad i = 1, 2.$$

Combine (43) with (41) and (40) one obtains

$$(44) \quad \text{Lip}((u_i)'') \leq \|\Phi''\|_{L^\infty} \|v_0'\|_{L^\infty} \leq 23(C_V^{3/2} + C_V).$$

We now compare the full discrete interaction argument $A_{n,m}^i[u_c]$ with its localized version. Define

$$(45) \quad B_{n,m}^i[u_c] := (-1)^{3-i} h n + (u_c^i)_n - (u_c^{3-i})_n - 2\pi m, \quad n \in I_i, \quad m \in \mathbb{Z},$$

where all indices are understood periodically, as before. Let

$$\rho_{n,m}^i := A_{n,m}^i[u_c] - B_{n,m}^i[u_c].$$

Using (13) and (45), we obtain

$$\begin{aligned} \rho_{n,m}^i &= (-1)^{3-i} \frac{h}{4\pi} \left((u_c^i)_n + (u_c^{3-i})_n \right) \\ &\quad - \left(1 + (-1)^i \frac{h}{4\pi} \right) \left((u_c^{3-i})_{n+m} - (u_c^{3-i})_n \right) - 2\pi (-1)^i \frac{h}{4\pi} m. \end{aligned}$$

Apply (42) to the sampled continuum pair u_c , we obtain

$$(46) \quad |(u_c^i)_{n+m} - (u_c^i)_n| \leq \|(u_i)'\|_{L^\infty} |x_{n+m}^i - x_n^i| \leq 3 \left(\sqrt{C_V} + 1 \right) h_i |m|, \quad i = 1, 2.$$

Therefore, combine (46) and (21), one obtains

$$(47) \quad \begin{aligned} |\rho_{n,m}^i| &\leq \frac{h}{2} + \left(1 + \frac{h}{4\pi} \right) 3 \left(\sqrt{C_V} + 1 \right) h_{3-i} |m| + \frac{h}{2} |m| \\ &\leq 7 \left(\sqrt{C_V} + 1 \right) (1 + |m|) h. \end{aligned}$$

We claim that

$$(48) \quad \left| 2 \sum_{m \in \mathbb{Z}} [V'(A_{n,m}^i[u_c]) - V'(B_{n,m}^i[u_c])] \right| \leq 1260 \left(C_V + C_V^{3/2} \right) h, \quad i = 1, 2.$$

Set

$$C_{n,m,\theta}^i := B_{n,m}^i[u_c] + \theta \rho_{n,m}^i = (1 - \theta) B_{n,m}^i[u_c] + \theta A_{n,m}^i[u_c],$$

then

$$|C_{n,m,\theta}^i| \geq \frac{4\pi}{3} |m| - 5\pi.$$

Note that here the bound we used is different from the one we used in the proof of Lemma 5. This is due to the fact that we are now considering the sampled continuum pair u_c , which has a cleaner uniform bound (21). For $|m| \geq 9$,

$$|C_{n,m,\theta}^i| \geq \frac{2\pi}{3} |m|.$$

The near-field part $|m| \leq 8$ and the far-field decay give the weighted estimate

$$\sup_{i,n} \sum_{m \in \mathbb{Z}} (1 + |m|) \sup_{\theta \in [0,1]} |V''(C_{n,m,\theta}^i)| \leq 90C_V.$$

It follows that, by the mean value theorem,

$$\begin{aligned} \left| 2 \sum_m [V'(A_{n,m}^i[u_c]) - V'(B_{n,m}^i[u_c])] \right| &\leq 2 \sum_m \sup_{\theta} |V''(C_{n,m,\theta}^i)| |\rho_{n,m}^i| \\ &\leq 1260 (C_V + C_V^{3/2}) h. \end{aligned}$$

For $n \in I_i$, set

$$s_n := hn + (u_c^1)_n - (u_c^2)_n.$$

Then

$$B_{n,m}^i[u_c] = (-1)^{3-i} s_n - 2\pi m.$$

Using the evenness of V , so that V' is odd, and applying the periodization (37), we obtain

$$2 \sum_{m \in \mathbb{Z}} V'(B_{n,m}^i[u_c]) = (-1)^{3-i} \Phi'(s_n).$$

Thus, for $i = 1, 2$, we have

$$(49) \quad |F_n^i(u_c) - (-1)^{3-i} \Phi'(s_n)| \leq 1260 (C_V + C_V^{3/2}) h.$$

It remains to estimate the phase error. Compare s_n with the continuum phase at the grid point x_n^i . Apply (42), we obtain

$$(50) \quad \begin{aligned} |s_n - v_0(x_n^i)| &\leq |h - h_i|n + |u_1(x_n^1) - u_1(x_n^i)| + |u_2(x_n^2) - u_2(x_n^i)| \\ &\leq \frac{h}{2} + \|(u_{3-i})'\|_{L^\infty} |x_n^{3-i} - x_n^i| \leq 4 (\sqrt{C_V} + 1) h. \end{aligned}$$

By (40), $\text{Lip}(\Phi') \leq 4C_V$, and hence

$$(51) \quad |\Phi'(s_n) - \Phi'(v_0(x_n^i))| \leq 16(C_V^{3/2} + C_V)h.$$

On the other hand, apply the central-difference expansion with integral remainder and combine with (44), we obtain

$$(52) \quad |\Delta_i u_c^i - (u_i)''(x_n^i)| \leq \frac{h_i}{3} \text{Lip}((u_i)'') \leq 12(C_V^{3/2} + C_V)h.$$

Combine (43), (49), (51), and (52), we obtain

$$\sup_{n,i} |r_n^i| \leq 1300(C_V^{3/2} + C_V)h.$$

This proves (38). Finally,

$$\|r\|_{\ell_h^2} = \sqrt{\sum_{i=1}^2 h_i \sum_{n \in I_i} |r_n^i|^2} \leq 5000(C_V^{3/2} + C_V)h.$$

□

We can now combine the consistency estimate with the stability estimate to obtain the main convergence result.

Theorem 5 (Discrete-to-continuum convergence). *For $N \geq 2$, let $V \in C^2(\mathbb{R})$ be an even potential satisfying the decay condition (14); let the continuum misfit potential Φ defined by (37) satisfy condition (2). Assume that there is a uniform gap $1 - C_P L_F > 0$. Then for u_d the symmetry-normalized discrete minimizer and u_c the sampled normalized continuum minimizer, we have the following convergence result in the discrete H^2 norm:*

$$(53) \quad \|u_d - u_c\|_{\ell_h^2} + \|\nabla_h(u_d - u_c)\|_{\ell_h^2} + \|\Delta_h(u_d - u_c)\|_{\ell_h^2} \leq C_{\text{conv}}(C_V, C_P, L_F) h,$$

equivalently,

$$\|u_d - u_c\|_{H_h^2} \leq C_{\text{conv}}(C_V, C_P, L_F) h,$$

where

$$(54) \quad C_{\text{conv}}(C_V, C_P, L_F) := \frac{5000}{1 - C_P L_F} \left[\left(1 + \sqrt{C_P} + C_P\right) \left(C_V + C_V^{3/2}\right) + \left(1 + \left(1 + \sqrt{C_P}\right) L_F\right) \right].$$

Proof. Let $w := u_d - u_c$. By Theorem 4,

$$\|r\|_{\ell_h^2} \leq 5000(C_V^{3/2} + C_V)h.$$

It remains to estimate the mean term in the stability bound.

Since the normalized continuum minimizer v_0 is odd and satisfies $v_0(x + 2\pi) = v_0(x) + 2\pi$, the associated pair

$$u_1(x) = \frac{v_0(x) - x}{2}, \quad u_2(x) = \frac{x - v_0(x)}{2}$$

are both odd and 2π -periodic, satisfying $u_i(2\pi - x) = u_i(-x) = -u_i(x)$. Hence, for $1 \leq n \leq N_i - 1$,

$$u_i(x_{N_i-n}^i) = u_i(2\pi - x_n^i) = -u_i(x_n^i).$$

The index $n = 0$ contributes $u_i(0) = 0$, and, when N_i is even, the additional fixed index $n = N_i/2$ contributes $u_i(\pi) = 0$. Therefore $(u_c^i) = 0$ for $i = 1, 2$.

For the symmetry-normalized discrete minimizer, by Corollary 1, we have

$$|(u_d^i)| \leq \frac{h}{4}, \quad i = 1, 2.$$

So for $w = u_d - u_c$,

$$|(w^1)| + |(w^2)| = |(u_d^1)| + |(u_d^2)| \leq \frac{h}{2}.$$

Applying the discrete stability estimate (26), we obtain

$$\|w\|_{\ell_h^2} + \|\nabla_h w\|_{\ell_h^2} + \|\Delta_h w\|_{\ell_h^2} \leq C_{\text{conv}}(C_V, C_P, L_F) h,$$

where C_{conv} is defined as in (54). $\square$

Using the discrete Sobolev embedding as in Appendix D, we can immediately obtain the following convergence result in the ℓ^∞ norm.

Corollary 2 (Convergence in the discrete maximum norm). *Under the assumptions of the convergence theorem 5, we have*

$$\|u_d - u_c\|_\infty \leq C_\infty(C_V, C_P, L_F) h,$$

where

$$C_\infty(C_V, C_P, L_F) := \frac{1}{4} + \sqrt{2\pi} C_{\text{conv}}(C_V, C_P, L_F)$$

Proof. Set $w := u_d - u_c$. For each layer i , following the proof as in Appendix D, the discrete 1D-Sobolev embedding gives

$$\|w^i\|_{\infty, i} \leq |(w^i)| + \sqrt{2\pi} \|\nabla_{h_i} w^i\|_{\ell_i^2}.$$

The sampled continuum means vanish exactly, while the normalized discrete minimizer satisfies $|(u_d^i)| \leq \frac{h}{4}$. Thus

$$|(w^i)| = |(u_d^i) - (u_c^i)| \leq \frac{h}{4}.$$

Taking the maximum over $i = 1, 2$,

$$\|w\|_{\infty} \leq \frac{h}{4} + \sqrt{2\pi} \|\nabla_h w\|_{\ell_h^2} \leq \left(\frac{1}{4} + \sqrt{2\pi} C_{\text{conv}} \right) h,$$

where the last inequality follows from the convergence theorem 5. This proves the result. $\square$

APPENDIX A. DERIVATION OF THE SIMPLIFIED DISCRETE MODEL

We now rewrite the energy (11) in a simplified, rescaled form (12). Start from intralayer terms, reindex the argument by $m' = m - n$ and rescale the displacement field as $\xi_n^i = \frac{h_i}{2\pi} u_n^i$. Assume there exists a rescaled 1D even potential $\tilde{V} : \mathbb{R} \rightarrow \mathbb{R}$ such that for both layers

$$2h_i^2 V_{i,i}(h_i s) = \tilde{V}(s), \quad \tilde{V}(-s) = \tilde{V}(s),$$

we get

$$E_{\text{intra}}^i[u] = \frac{1}{2h_i} \sum_{n=0}^{N_i-1} \sum_{m' \in \mathbb{Z}} \tilde{V}\left(\frac{u_n^i - u_{n+m'}^i}{2\pi} - m'\right)$$

Now assume the unperturbed state is a nondegenerate minimum, which gives $\tilde{V}'(-m') = 0$. Expand around this minimum and impose the nearest-neighbor truncation, up to an additive constant and a scaling factor, we obtain

$$E_{\text{intra}}^i[u] = \frac{h_i}{2} \sum_n \left(\frac{u_{n+1}^i - u_n^i}{h_i} \right)^2.$$

For the potential terms, we first assume the pair potentials are symmetric and even, i.e. $V^{12} = V^{21} = \bar{V}$. Reindex the argument by $m' = m - n$ and rescale the displacement field as $\xi_n^i = \frac{h_i}{2\pi} u_n^i$, we obtain for $i, 3-i \in \{1, 2\}$, $i \neq 3-i$

$$E_{\text{inter}}^i = h_i \sum_{n=0}^{N_i-1} \sum_{m' \in \mathbb{Z}} \bar{V}\left((h_i - h_{3-i})n + \frac{h_i}{2\pi} u_n^i - \frac{h_{3-i}}{2\pi} u_{n+m'}^{3-i} - h_{3-i} m'\right),$$

Rescale the interlayer potential as $\bar{V}(x) = V(\frac{4\pi}{h_1+h_2}x)$ and denote the harmonic mean $h = \frac{2h_1 h_2}{h_1 + h_2}$, we obtain

$$E_{\text{inter}}^i = h_i \sum_{n=0}^{N_i-1} \sum_{m' \in \mathbb{Z}} V\left((-1)^{3-i} h n + \frac{h}{h_{3-i}} u_n^i - \frac{h}{h_i} u_{n+m'}^{3-i} - 2\pi \frac{h}{h_i} m'\right).$$

APPENDIX B. PROOF OF COROLLARY 1

Proof. Choose $c = c(d)$ so that the transformed means sum to zero:

$$(\tilde{u}^1) + (\tilde{u}^2) = 0.$$

Since

$$(\tilde{u}^1) + (\tilde{u}^2) = (u^1) + (u^2) + h_2 d + (h_1 + h_2) c,$$

take

$$c(d) := -\frac{(u^1) + (u^2) + h_2 d}{h_1 + h_2}.$$

Then indeed

$$(\tilde{u}^1) + (\tilde{u}^2) = 0.$$

Substitute this into $(\tilde{u}^1)$:

$$(\tilde{u}^1)(d) = (u^1) + h_2 d - h_2 \frac{(u^1) + (u^2) + h_2 d}{h_1 + h_2} = \frac{h_1 (u^1) - h_2 (u^2) + h_1 h_2 d}{h_1 + h_2}.$$

Choose ℓ_* to be the nearest integer to

$$\frac{h_2 (u^2) - h_1 (u^1)}{h_1 h_2}.$$

Then

$$\left| \frac{h_2(u^2) - h_1(u^1)}{h_1 h_2} - \ell_* \right| \leq \frac{1}{2},$$

which gives

$$|(\tilde{u}^1)(\ell_*)| \leq \frac{h_1 h_2}{2(h_1 + h_2)}.$$

Since $(\tilde{u}^2) = -(\tilde{u}^1)$, the same bound holds for layer 2. Finally,

$$\frac{h_1 h_2}{2(h_1 + h_2)} = \frac{1}{4} \frac{2h_1 h_2}{h_1 + h_2} = \frac{h}{4}.$$

□

APPENDIX C. PROOF OF LEMMA 2

Proof. Note that by (13), we have the estimate

$$|A_{n,m}^i[u]| \geq \pi|m| - 2\pi(1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}).$$

We split the sum into a near field and a far field. Let

$$M := \left\lceil 4(1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}) \right\rceil.$$

For $|m| \leq M$, we only use the bound $|G| \leq C_V$, then

$$\sum_{|m| \leq M} |G(A_{n,m}^i[u])| \leq C_V(2M + 1) \leq 11C_V \left(1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}\right)$$

For $|m| > M$, we have

$$\pi|m| - 2\pi(1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}) \geq \frac{\pi}{2}|m|,$$

and hence for any $n \in I_i$, since $r > 2$,

$$\sum_{|m| > M} |G(A_{n,m}^i[u])| \leq C_V \sum_{|m| > M} \left(1 + \frac{\pi}{2}|m|\right)^{-r} \leq 2C_V \sum_{m=1}^{\infty} \left(1 + \frac{\pi}{2}m\right)^{-r} \leq 2C_V.$$

Combining the near-field and far-field estimates gives

$$\sum_{m \in \mathbb{Z}} \sup_{n \in I_i} |G(A_{n,m}^i[u])| \leq 13C_V \left(1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}\right).$$

□

APPENDIX D. PROOF OF LEMMA 3

Following the Sobolev–Morrey embedding as in [8, §5.6.2, Thm. 5] and applying the 1D finite-difference analogue as in [8, §5.10, Ex. 4], we have for any indices j, k ,

$$|u_k^i - u_j^i| \leq \sqrt{2\pi} \|\nabla_i^+ u^i\|_{\ell_i^2}.$$

Then, since the layer mean lies between the minimum and maximum values,

$$\|u^i - (u^i)\|_{\infty,i} \leq \sqrt{2\pi} \|\nabla_i^+ u^i\|_{\ell_i^2}.$$

With the mean-fixing normalization in lemma 1, we have

$$\|u^1\|_{\infty,1} + \|u^2\|_{\infty,2} \leq 2\sqrt{\pi} \sqrt{G[u]} + \frac{h}{2},$$

where

$$G[u] := \|\nabla_{h_1}^+ u^1\|_{\ell_1^2}^2 + \|\nabla_{h_2}^+ u^2\|_{\ell_2^2}^2.$$

Then Lemma 2 gives the interaction lower bound

$$\sum_{i=1}^2 \sum_{n \in I_i} \sum_{m \in \mathbb{Z}} V(A_{n,m}^i[u]) \geq -52\pi C_V (1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}).$$

hence

$$E_N[u] \geq \frac{1}{2}G[u] - 52\pi C_V \left(1 + \frac{h}{2} + 2\sqrt{\pi}\sqrt{G[u]}\right)$$

Since $E_N[u] \leq C_0 \leq 52\pi C_V$, it follows that $G[u]$ is uniformly bounded, and one can calculate that the sum of the layerwise supremum norms is bounded by

$$\|u^1\|_{\infty,1} + \|u^2\|_{\infty,2} \leq \frac{h}{2} + 42\pi\sqrt{C_V} + 416\pi^2 C_V.$$

APPENDIX E. PROOF OF LEMMA 4

Proof. By lemma 2, applied with V' in place of G , the discrete maximum-norm bound on u implies

$$\sum_{m \in \mathbb{Z}} \sup_{n \in I_i} |V'(A_{n,m}^i[u])| \leq 13C_V (1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}).$$

Therefore, using the Euler–Lagrange equation (17),

$$\|\Delta_i u^i\|_{\infty,i} = \|F^i(u)\|_{\infty,i} \leq 26C_V (1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}).$$

Then w.l.o.g. for any indices n, m within a period, we have

$$|\nabla_i^+ u_n^i - \nabla_i^+ u_m^i| \leq h_i |n - m| \|\Delta_i u^i\|_{\infty,i} \leq 2\pi \|\Delta_i u^i\|_{\infty,i}.$$

Moreover, periodicity gives

$$\sum_{n \in I_i} \nabla_i^+ u_n^i = \sum_{n \in I_i} \frac{u_{n+1}^i - u_n^i}{h_i} = 0.$$

Thus, unless every $\nabla_i^+ u_n^i = 0$, for each n there exists some m such that $\nabla_i^+ u_n^i$ and $\nabla_i^+ u_m^i$ have opposite signs. Consequently,

$$|\nabla_i^+ u_n^i| \leq |\nabla_i^+ u_n^i - \nabla_i^+ u_m^i| \leq 2\pi \|\Delta_i u^i\|_{\infty,i}.$$

Therefore,

$$\|\nabla_i^+ u^i\|_{\infty,i} \leq 2\pi \|\Delta_i u^i\|_{\infty,i} \leq 52\pi C_V (1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}).$$

Finally, by telescoping and applying Lemma 3, we have

$$\begin{aligned} |u_{n+m}^i - u_n^i| &\leq \sum_{r=0}^{|m|-1} h_i \|\nabla_i^+ u^i\|_{\infty,i} \leq 52\pi C_V (1 + \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}) |m| h_i \\ &\leq 32448\pi^3 (C_V + C_V^{3/2} + C_V^2) |m| h. \end{aligned}$$

□

APPENDIX F. PROOF OF LEMMA 5

Proof. For simplicity, denote only in this proof

$$\Lambda_V := 32448\pi^3 \left(C_V + C_V^{3/2} + C_V^2 \right),$$

$$U := \|u^1\|_{\infty,1} + \|u^2\|_{\infty,2}.$$

Recall that $A_{n,m}^i[u]$ is defined as

$$A_{n,m}^i[u] = (-1)^{3-i}hn + \frac{h}{h_{3-i}}u_n^i - \frac{h}{h_i}u_{n+m}^{3-i} - 2\pi\frac{h}{h_i}m, \quad n \in I_i.$$

Now for (18), define

$$B_{n,m}^i[u] := (-1)^{3-i}hn + u_n^i - u_{n+m}^{3-i} - 2\pi m, \quad n \in I_i.$$

Then

$$E^i[u] - E_{\text{loc}}^i[u] = \sum_{n \in I_i} \sum_{m \in \mathbb{Z}} (V(A_{n,m}^i) - V(B_{n,m}^i)).$$

With $\frac{h_1-h_2}{h_1+h_2} = \frac{h}{4\pi}$, one can estimate

$$|A_{n,m}^i - B_{n,m}^i| \leq \frac{h}{4\pi}U + \left(1 + \frac{h}{4\pi}\right) |u_{n+m}^{3-i} - u_n^{3-i}| + \frac{h}{2}|m|.$$

Note that by Lemma 4, for all $n \in I_i$, $m \in \mathbb{Z}$,

$$|u_{n+m}^i - u_n^i| \leq \Lambda_V |m|h,$$

Then

$$(55) \quad |A_{n,m}^i - B_{n,m}^i| \leq h \left[\frac{U}{4\pi} + \left(\frac{4}{3}\Lambda_V + \frac{1}{2} \right) |m| \right].$$

Fix n , by mean value theorem,

$$(56) \quad |V(A_{n,m}^i) - V(B_{n,m}^i)| \leq \sup_{0 \leq \theta \leq 1} |V'(C_{n,m,\theta}^i)| |A_{n,m}^i - B_{n,m}^i|,$$

where

$$C_{n,m,\theta}^i := B_{n,m}^i + \theta(A_{n,m}^i - B_{n,m}^i), \quad 0 \leq \theta \leq 1.$$

The same lower-bound argument as in Lemma 2 gives

$$|C_{n,m,\theta}^i| \geq \pi|m| - 2\pi(1+U),$$

uniformly in

$$i = 1, 2, \quad n \in I_i, \quad m \in \mathbb{Z}, \quad \theta \in [0, 1].$$

Choose $M = \lceil 4(1+U) \rceil$. Now apply Lemma 2 to V' , we retain

$$(57) \quad \sum_{m \in \mathbb{Z}} \sup_{\substack{n \in I_i \\ 0 \leq \theta \leq 1}} |V'(C_{n,m,\theta}^i)| \leq 13C_V(1+U).$$

For the $|m|$ -weighted sum, we retain

$$(58) \quad \sum_{m \in \mathbb{Z}} |m| \sup_{\substack{n \in I_i \\ 0 \leq \theta \leq 1}} |V'(C_{n,m,\theta}^i)| \leq 31C_V(1+U)^2.$$

Now we can combine (55), (56), (57) and (58) to obtain

$$|E^i[u] - E_{\text{loc}}^i[u]| \leq 2\pi h \left[\frac{U}{4\pi} (13C_V(1+U)) + \left(\frac{4}{3}\Lambda_V + \frac{1}{2} \right) (31C_V(1+U)^2) \right].$$

To simplify the above estimate, use $U(1+U) \leq (1+U)^2$, plug in the definition of Λ_V and relax the constant terms, we obtain

$$|E^i[u] - E_{\text{loc}}^i[u]| \leq hC_V(1+U)^2 \left[34\pi + 2682368\pi^4 \left(C_V + C_V^{3/2} + C_V^2 \right) \right].$$

Next, Lemma 3 gives

$$U \leq \frac{h}{2} + 42\pi\sqrt{C_V} + 416\pi^2 C_V$$

Then

$$|E^i[u] - E_{\text{loc}}^i[u]| \leq h\mathcal{P}(C_V).$$

Combine this with the symmetry (19) of $E_{\text{loc}}^i[u]$, we see that $E^i[u]$ also enjoys (19) approximately, in the sense that

$$|E^i[\tau_k u] - E^i[u]| \lesssim h.$$

□

APPENDIX G. PROOF OF LEMMA 6

Proof. Recall the uniform bound for u_c

$$\|u_c^1\|_{\infty,1} + \|u_c^2\|_{\infty,2} \leq 2\pi.$$

and for u_d

$$\|u_d^1\|_{\infty,1} + \|u_d^2\|_{\infty,2} \leq 42\pi\sqrt{C_V} + 416\pi^2 C_V + \frac{2\pi}{3}.$$

Then for any $t \in [0, 1]$,

$$\begin{aligned} \|z_t^1\|_{\infty,1} + \|z_t^2\|_{\infty,2} &\leq (1-t) \left(\|u_c^1\|_{\infty,1} + \|u_c^2\|_{\infty,2} \right) + t \left(\|u_d^1\|_{\infty,1} + \|u_d^2\|_{\infty,2} \right) \\ &\leq \frac{8\pi}{3} + 42\pi\sqrt{C_V} + 416\pi^2 C_V. \end{aligned}$$

We next estimate the V'' -sum uniformly in t . The lower-bound calculation used in the proof of Lemma 2 gives

$$|A_{n,m}^i[z_t]| \geq \pi|m| - 2\pi \left(1 + \|z_t^1\|_{\ell_1^\infty} + \|z_t^2\|_{\ell_2^\infty} \right).$$

Denote only in this proof

$$\mathcal{M}_V := \max_{i=1,2} \sup_{N \in \mathbb{N}} \sup_{n \in I_i} \sum_{m \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{n,m}^i[z_t])|,$$

then by the same argument as in Lemma 2, we have

$$\begin{aligned} \mathcal{M}_V &\leq 13C_V \left(1 + \|z_t^1\|_{\infty,1} + \|z_t^2\|_{\infty,2} \right) \\ &\leq 13C_V \left(1 + \frac{8\pi}{3} + 42\pi\sqrt{C_V} + 416\pi^2 C_V \right) \end{aligned}$$

It remains to prove the force estimate. For each fixed n , apply the mean value theorem on (16), we have

$$|F_n^i(u_d) - F_n^i(u_c)| \leq 2 \sum_{m \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{n,m}^i[z_t])| \left(\frac{h}{h_{3-i}} |w_n^i| + \frac{h}{h_i} |w_{n+m}^{3-i}| \right)$$

For the term with w_n^i , we can directly use the assumption (27) to bound it by $2\frac{h}{h_{3-i}}M_V|w_n^i|$. Square the above inequality, and use Cauchy–Schwarz for the w_{n+m}^{3-i} term, we have

$$\begin{aligned} & |F_n^i(u_d) - F_n^i(u_c)|^2 \\ & \leq 8\left(\frac{h}{h_{3-i}}\right)^2 M_V^2 |w_n^i|^2 + 8\left(\frac{h}{h_i}\right)^2 M_V \left(\sum_{m \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{n,m}^i[z_t])| |w_{n+m}^{3-i}|^2 \right). \end{aligned}$$

Multiply both sides by h_i and sum over $n \in I_i$, we have

$$\begin{aligned} & \|F^i(u_d) - F^i(u_c)\|_{\ell_i^2}^2 \leq 8\left(\frac{h}{h_{3-i}}\right)^2 M_V^2 \|w^i\|_{\ell_i^2}^2 \\ & + 8\left(\frac{h}{h_i}\right)^2 M_V h_i \sum_{n \in I_i} \sum_{m \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{n,m}^i[z_t])| |w_{n+m}^{3-i}|^2. \end{aligned}$$

Note that for fixed $n \in I_i$, $p \in I_{3-i}$ and for any $q \in \mathbb{Z}$, the terms $A_{n,m}^i$ are related by

$$A_{n,p-n+qN_{3-i}}^i = -A_{p,n-p-qN_i}^{3-i}.$$

Take $m = p - n + qN_{3-i}$ for some $q \in \mathbb{Z}$. For fixed n , the mapping

$$m := p - n + qN_{3-i} \in \mathbb{Z} \mapsto (p, q) \in I_{3-i} \times \mathbb{Z}$$

is a bijection. Therefore, we can rewrite the second term as

$$8\left(\frac{h}{h_i}\right)^2 M_V h_i \sum_{n \in I_i} \sum_{p \in I_{3-i}} \sum_{q \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{n,p-n+qN_{3-i}}^i[z_t])| |w_p^{3-i}|^2.$$

Switch the order of summation, use the above relation between $A_{n,m}^i$ and the evenness of V'' gives

$$8\left(\frac{h}{h_i}\right)^2 M_V h_i \sum_{p \in I_{3-i}} \sum_{n \in I_i} \sum_{q \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{p,n-p-qN_i}^{3-i}[z_t])| |w_p^{3-i}|^2.$$

Now for each fixed $p \in I_{3-i}$, the mapping

$$(n, q) \in I_i \times \mathbb{Z} \mapsto m' := n - p - qN_i \in \mathbb{Z}$$

is a bijection, thus we can rewrite the second term as

$$8\left(\frac{h}{h_i}\right)^2 M_V h_i \sum_{p \in I_{3-i}} \sum_{m \in \mathbb{Z}} \sup_{t \in [0,1]} |V''(A_{p,m'}^{3-i}[z_t])| |w_p^{3-i}|^2.$$

Now we can use the assumption (27) to bound it by

$$8\left(\frac{h}{h_i}\right)^2 \left(\frac{h_i}{h_{3-i}}\right) M_V^2 \|w^{3-i}\|_{\ell_{3-i}^2}^2.$$

Combine the two terms, we have

$$\|F^i(u_d) - F^i(u_c)\|_{\ell_i^2}^2 \leq 8\left(\frac{h}{h_{3-i}}\right)^2 M_V^2 \|w^i\|_{\ell_i^2}^2 + 8\left(\frac{h}{h_i}\right)^2 \left(\frac{h_i}{h_{3-i}}\right) M_V^2 \|w^{3-i}\|_{\ell_{3-i}^2}^2.$$

Combine the two layers estimates,

$$\|F(u_d) - F(u_c)\|_{\ell_h^2}^2 \leq 32\mathcal{M}_V^2 \left(\|w^1\|_{\ell_1^2}^2 + \|w^2\|_{\ell_2^2}^2 \right).$$

Therefore,

$$\|F(u_d) - F(u_c)\|_{\ell_h^2} \leq 4\sqrt{2}\mathcal{M}_V \|u_d - u_c\|_{\ell_h^2} = L_F(C_V) \|u_d - u_c\|_{\ell_h^2},$$

where $L_F(C_V) := 4\sqrt{2}\mathcal{M}_V$ is a constant depending only on C_V . In particular, we have the estimate

$$L_F(C_V) \leq 21632\sqrt{2}\pi^2 \left(C_V + C_V^{3/2} + C_V^2 \right).$$

□

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